Introduction to Online Learning

Lecture notes — summary of the fundamentals and Follow the Leader

1 From Statistical Learning to Online Learning

1.1 Parametric models and classical learning

In classical supervised learning, we observe a sample $(X_i, Y_i)_{i=1}^n \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{(X,Y)}$ and estimate a parameter $\theta \in \Theta$ in order to predict Y from X.

Definition 1.1 (Parametric model)**.** A family of predictors is parametric when it can be written as $\mathcal{M} = \{\mu_\theta : \mathcal{X} \to \mathcal{Y} ; \theta \in \Theta\}$, where Θ is finite-dimensional. Learning amounts to estimating $\widehat{\theta}$ from the observations.

The usual examples all fall within this framework, even when they are nonlinear:

Model	Mathematical formulation / purpose
Linear regression	$Y = \beta_0 + \beta^\top X + \varepsilon$; predict a numerical quantity.
Logistic regression	$\mathbb{P}(Y = 1 \mid X = x) = \left[1 + \exp(-\beta_0 - \beta^\top x)\right]^{-1}$; classify.
Generalized linear model	$g(\mathbb{E}[Y \mid X = x]) = \beta_0 + \beta^\top x$.
Neural network	$\mu_\theta = h_L \circ \cdots \circ h_1$; weights and biases form θ.
ARIMA	Parametric time-series model incorporating dependencies and differencing.

A **time series** is an observed realization of a stochastic process $(Y_t)_{t \in \mathcal{T}}$. Daily bread sales or hourly temperatures are two examples. Unlike an i.i.d. sample, observations may depend on the past:

$$Y_t = \underbrace{2 + 0.1 Y_{t-1}}_{\text{predictable component}} + \underbrace{\varepsilon_t}_{\text{innovation}}, \qquad \mathbb{E}[\varepsilon_t] = 0, \quad \mathrm{Var}(\varepsilon_t) = \sigma^2.$$

If the innovation is independent of the past, $\mathbb{E}[Y_t \mid Y_{t-1}] = 2 + 0.1 Y_{t-1}$ and $\mathrm{Var}(Y_t \mid Y_{t-1}) = \sigma^2$. This is an AR(1) model.

Remark 1.2. Time indexing does not imply dependence: i.i.d. white noise is also a time series. Nevertheless, a model should distinguish the predictable structure from residual uncertainty.

Remark 1.3 (Two paradigms, two kinds of guarantees). In the i.i.d. statistical framework, randomness is modeled through a distribution $\mathbb{P}_{(X,Y)}$ and we study the properties of an estimator or predictor, including its out-of-sample risk. In the online protocol, the analysis may concern an *arbitrary* sequence of losses $\ell_1, \ldots, \ell_T$, without assuming a stationary underlying distribution. The performance criterion is then **regret**. These frameworks are not mutually exclusive: one may learn online from stochastic data, but the assumptions and objectives differ.

1.2 Loss function and statistical optimization

Selecting a model is not enough: we also need a *criterion* to measure its errors. For simple linear regression $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, the residual is $r_i(\beta) = Y_i - \beta_0 - \beta_1 X_i$. Under least squares,

$$\mathcal{L}_n(\beta) = \frac{1}{n} \sum_{i=1}^n r_i(\beta)^2, \qquad \widehat{\beta} \in \arg\min_{\beta \in \mathbb{R}^2} \mathcal{L}_n(\beta).$$

The **empirical risk** $\mathcal{L}_n$ can be computed from the data; the **population risk** $\mathcal{R}(\beta) = \mathbb{E}[\ell(\beta; X, Y)]$ depends on the data distribution. Statistical Learning studies, among other things, the relationship between them.

Definition 1.4 (Gradient descent)**.** For a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$, the vector $\nabla f(\theta)$ gives the direction of greatest local increase when it is nonzero. Gradient descent follows

$$\boxed{\theta^{(k+1)} = \theta^{(k)} - \eta_k \nabla f(\theta^{(k)}), \qquad \eta_k > 0.}$$

The negative gradient is a direction of *local* decrease, not necessarily the direction pointing towards a global minimum.

For simple linear regression, the gradient is explicit:

$$\nabla \mathcal{L}_n(\beta) = -\frac{2}{n} \begin{pmatrix} \sum_{i=1}^n r_i(\beta) \\ \sum_{i=1}^n X_i r_i(\beta) \end{pmatrix}.$$

Convexity of the squared loss provides additional convergence guarantees when an appropriate step size is chosen.

Example 1.5 (One gradient descent iteration)**.** With $J(z) = \frac{1}{2}(z-15)^2$, we have $J'(z) = z-15$. Starting from $z^{(0)} = 10$ with step size $\eta = 0.2$,

$$z^{(1)} = 10 - 0.2(10 - 15) = 11, \qquad J(10) = 12.5, \quad J(11) = 8.$$

The updated parameter reduces the loss. Unlike FTL, gradient descent here *iteratively* optimizes the same objective, fixed in advance.

1.3 Why the loss function matters

The loss function specifies which prediction errors are costly. It does not automatically follow from the predictor architecture. For a scalar prediction $\widehat{y}$ and observation y, common choices lead to different objectives:

Loss	Expression	Interpretation
Squared	$(y - \widehat{y})^2$	Large residuals receive a strong penalty.
Absolute	$\lvert y - \widehat{y}\rvert$	Linear penalty for prediction errors.
Asymmetric	$c_-(y - \widehat{y})_+ + c_+(\widehat{y} - y)_+$	Understock and overstock incur different costs.

In the third case, $c_- > 0$ is the unit cost of unmet demand and $c_+ > 0$ is the cost of one excess unit ordered. This loss is more natural than a symmetric loss for an inventory decision. We will use the first two losses as our mathematical examples.

1.4 The empirical mean: a first reference optimum

Given real observations $x_1, \dots, x_n$, consider the best *constant* under squared loss.

Theorem 1.6 (Mean and least squares)**.** *The minimizer is unique and equals the empirical mean:*

$$\boxed{z^\star = \arg\min_{z \in \mathbb{R}} \frac{1}{n} \sum_{i=1}^n (x_i - z)^2 = \frac{1}{n} \sum_{i=1}^n x_i.}$$

Proof. For $J(z) = n^{-1} \sum_i (x_i - z)^2$, $J'(z) = 2(z - n^{-1} \sum_i x_i)$ and $J''(z) = 2 > 0$. Therefore, the unique stationary point is the unique global minimum. □

For instance, the observations $(2, 3, 10)$ give $z^\star = 5$. This result will help us understand the FTL algorithm.

1.5 The online protocol and regret

In Online Learning, the learner sequentially chooses decisions $\theta_t \in \mathcal{K}$; the loss ℓ_t is revealed only after the decision at time t. Evaluation concerns the total loss incurred, not the quality of an estimator computed once from a complete sample.

Definition 1.7 (Hindsight comparator and regret)**.** The best *fixed* decision chosen in hindsight is $\theta^\star \in \arg\min_{\theta\in\mathcal{K}} \sum_{t=1}^T \ell_t(\theta)$. Regret against a comparator $u \in \mathcal{K}$ and standard regret are

$$\mathrm{Reg}_T(u) = \sum_{t=1}^T [\ell_t(\theta_t) - \ell_t(u)], \quad \mathrm{Reg}_T = \sum_{t=1}^T \ell_t(\theta_t) - \min_{u\in\mathcal{K}} \sum_{t=1}^T \ell_t(u).$$

The quantity Reg_T/T is called *average regret*.

With the loss $\ell_t(\theta) = (Y_t - \theta)^2$, the optimal fixed comparator is $\theta^\star = T^{-1}\sum_{t=1}^T Y_t$: it knows all the past *and* the future, unlike the learner. Regret can be negative even against the best fixed comparator: a sequence of adaptive decisions can sometimes outperform every constant. For FTL with exact minimizers, however, we will prove a special nonnegativity property.

Remark 1.8 (Decisions and information: timeline). The order of events is crucial:

$$\underbrace{\theta_t \text{ chosen}}_{\text{before observation}} \longrightarrow \underbrace{\ell_t \text{ revealed}}_{\text{loss } \ell_t(\theta_t)} \longrightarrow \underbrace{\theta_{t+1} \text{ updated}}_{\text{for the next round}}.$$

This information constraint is precisely what distinguishes an online policy from optimization performed in hindsight.

2 Convexity, Subgradients and Online Optimization

2.1 From derivatives to the subdifferential

Some convex losses (notably absolute value) are not differentiable everywhere. Subgradients generalize gradients to this setting.

Definition 2.1 (Subgradient)**.** For a convex function $f : \mathbb{R}^d \to \mathbb{R}$, a vector $g \in \mathbb{R}^d$ is a subgradient of f at u if

$$\boxed{f(v) \geq f(u) + \langle g, v - u\rangle \quad \text{for all } v \in \mathbb{R}^d.}$$

The **subdifferential** is the set $\partial f(u) = \{g : \langle g, v-u\rangle \leq f(v) - f(u),\ \forall v\}$. If f is differentiable at u, then $\partial f(u) = \{\nabla f(u)\}$.

Geometrically, $v \mapsto f(u) + \langle g, v - u\rangle$ is a *global affine lower bound* for f that touches its graph at u. The inequality can be rearranged into a form useful for regret analysis:

$$\langle g, u - v\rangle \geq f(u) - f(v).$$

2.2 Absolute value and supporting lines

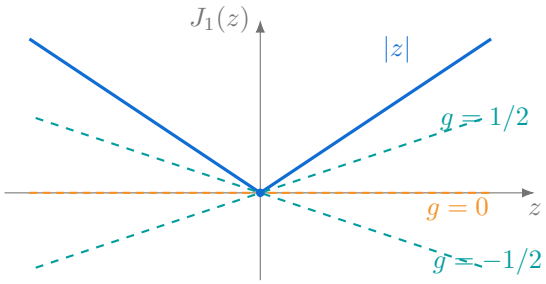
For $J_2(z) = z^2$, the minimum is at zero: $J_2'(z) = 2z$. By contrast, $J_1(z) = |z|$ has a kink at zero: $J_1'(z) = -1$ for $z < 0$ and $J_1'(z) = 1$ for $z > 0$. Let us compute its subdifferential at zero:

$$\partial J_1(0) = \{g \in \mathbb{R} : gv \leq |v|,\ \forall v \in \mathbb{R}\}.$$

For $v > 0$, we obtain $g \leq 1$; for $v < 0$, dividing by a negative number reverses the inequality and gives $g \geq -1$; for $v = 0$, there is no additional restriction. Conversely, these conditions suffice.

Hence,

$$\boxed{\partial|z| = \begin{cases} \{-1\}, & z < 0, \\ [-1,1], & z = 0, \\ \{1\}, & z > 0. \end{cases}}$$



Each dashed line touches $|z|$ at zero and remains below its graph: its slope belongs to $[-1, 1]$. The horizontal supporting line corresponds to $g = 0$.

Theorem 2.2 (Convex optimality condition)**.** *For a convex function $f : \mathbb{R}^d \to \mathbb{R}$,*

$$\boxed{u^\star \in \arg\min_{u \in \mathbb{R}^d} f(u) \iff 0 \in \partial f(u^\star).}$$

Proof. If $0 \in \partial f(u^\star)$, then, for all v, $f(v) \geq f(u^\star) + \langle 0, v - u^\star \rangle = f(u^\star)$. Conversely, if $u^\star$ minimizes f, this same inequality holds with $g = 0$. □

Remark 2.3. For scalar observations, squared loss $\sum_i (x_i - z)^2$ is minimized by the **mean**, whereas absolute loss $\sum_i |x_i - z|$ is minimized by a **median**. For $(2, 3, 10)$, the two answers are 5 and 3, respectively.

Example 2.4 (Another optimum: the median)**.** For $x_1 \leq \cdots \leq x_n$, consider $A(z) = \sum_i |x_i - z|$. Between two successive observations, the slope of A equals the number of observations to the left of z minus the number to its right. It is negative before the median and positive after it. Therefore $0 \in \partial A(z)$ exactly for the empirical medians; if n is even, the entire interval $[x_{n/2}, x_{n/2+1}]$ minimizes A.
This result illustrates a general principle: changing the loss changes the optimal parameter, and therefore the learning target.

2.3 Online Gradient Descent: an overview

The OGD algorithm updates a decision *after* the current loss is revealed, using a subgradient of that loss alone.

Definition 2.5 (Online Gradient Descent (OGD))**.** If $\mathcal{K} \subset \mathbb{R}^d$ is nonempty, closed, and convex, $g_t \in \partial \ell_t(\theta_t)$, and $\eta_t > 0$, then

$$\boxed{\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t),}$$

where $\Pi_{\mathcal{K}}$ denotes Euclidean projection.

Offline gradient descent uses the gradient of an objective fixed by the sample; OGD updates decisions sequentially and seeks a guarantee on cumulative losses.

Remark 2.6 (Why use subgradients?). In many problems (robust regression with absolute loss, quantile loss, or constrained optimization), the objective is convex but has nondifferentiable points. The subgradient inequality then replaces classical differential arguments and makes it possible to analyze algorithms such as OGD.

3 Asymptotic Review: Little o

The notation o formalizes the idea of a quantity being *negligible* relative to another. In the Taylor–Young formula,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o((x-a)^n), \qquad x \to a,$$

the remainder becomes negligible compared with $(x-a)^n$.

Definition 3.1 (Little o)**.** Provided $g(x) \neq 0$ in a punctured neighborhood of a,

$$\boxed{f(x) = o(g(x)) \ (x \to a) \iff \lim_{x \to a} \frac{f(x)}{g(x)} = 0.}$$

For example, $x^2 = o(x)$ as $x \to 0$, $x^2 = o(\sqrt{x})$ as $x \to 0^+$, whereas $x = o(x^2)$ as $x \to +\infty$. The limit must therefore be specified. For time-indexed sequences,

$$\frac{\sqrt{T}}{T} = \frac{1}{\sqrt{T}} \longrightarrow 0, \qquad \frac{\log T}{T} \longrightarrow 0 \qquad (T \to \infty).$$

Example 3.2 (Taylor remainder and asymptotic scales)**.** Near zero, the expansion $e^x = 1 + x + \frac{1}{2}x^2 + o(x^2)$ means that the error $e^x - (1 + x + \frac{1}{2}x^2)$ is small relative to x^2. Do not confuse O with o: writing $a_T = O(\sqrt{T})$ means that some $C > 0$ satisfies $|a_T| \leq C\sqrt{T}$ for sufficiently large T. This implies $a_T = o(T)$, but does not imply $a_T = o(\sqrt{T})$.

Proposition 3.3 (Sublinear regret)**.** *If* $\mathrm{Reg}_T = o(T)$*, then* $\mathrm{Reg}_T/T \to 0$*: average regret tends to zero. The bound* $|\mathrm{Reg}_T| \leq C\sqrt{T}$ *ensures this. An upper bound alone,* $\mathrm{Reg}_T \leq C\sqrt{T}$*, guarantees* $\limsup_T \mathrm{Reg}_T/T \leq 0$*, but not necessarily convergence to zero for regret against an arbitrary comparator.*

For FTL with exact minimizers, a telescoping identity will establish $\mathrm{Reg}_T \geq 0$. In that case, an $o(T)$ upper bound suffices to prove that average regret converges to zero. This conclusion does not automatically apply to every algorithm.

Regret growth	Average regret	Sublinear?
$C \log T$	$C \log(T)/T \to 0$	yes
$C\sqrt{T}$	$C/\sqrt{T} \to 0$	yes
CT	C	no

The asymptotic goal does not require bounded cumulative regret: regret can diverge to infinity and still be negligible on average.

4 Follow the Leader: Method and Example

4.1 The principle of hindsight decisions

Every day, we aim to predict the demand y_t for potatoes. The decision θ_t is made before y_t is revealed. Our pedagogical model uses real-valued quantities over $\mathcal{K} = [0, 20]$ and squared loss

$$\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2.$$

The factor $1/2$ makes differentiation convenient without changing the minimizers. In a real inventory problem, stockout and overstock costs are generally asymmetric; squared loss is a deliberate simplification here.

Definition 4.1 (Follow the Leader (FTL))**.** Initialize $\theta_1 \in \mathcal{K}$. For every $t \geq 2$,

$$\theta_t \in \arg\min_{\theta \in \mathcal{K}} \sum_{s=1}^{t-1} \ell_s(\theta).$$

The current decision minimizes *past* losses: it does not use the loss ℓ_t, which has not yet been revealed.

We must distinguish $\sum_{s<t} \ell_s(\theta)$, which evaluates the *same* decision over past rounds, from $\sum_{s \leq t} \ell_s(\theta_s)$, the loss actually incurred by a learner who may change its decision every day.

4.2 Numerical example over three days

Start with $\theta_1 = 10$, and let the actual demand be $y_1 = 15$. The loss is $\ell_1(10) = \frac{1}{2}(10-15)^2 = 12.5$. On day two, FTL solves

$$\theta_2 = \arg\min_{\theta \in [0,20]} \frac{1}{2}(\theta - 15)^2 = 15.$$

Then $y_2 = 12$, giving the loss $\ell_2(15) = \frac{1}{2}(15-12)^2 = 4.5$. On day three, the two observations give

$$\theta_3 = \arg\min_{\theta \in [0,20]} \left\{ \frac{1}{2}(\theta - 15)^2 + \frac{1}{2}(\theta - 12)^2 \right\}.$$

The derivative $2\theta - 27$ vanishes at $\theta_3 = 13.5$. We then observe $y_3 = 16$, and the loss is $\ell_3(13.5) = \frac{1}{2}(13.5-16)^2 = 3.125$.

t	θ_t	y_t	$\theta_t - y_t$	$\ell_t(\theta_t)$	$\sum_{s \leq t} \ell_s(\theta_s)$
1	10	15	-5	12.5	12.5
2	15	12	3	4.5	17
3	13.5	16	-2.5	3.125	20.125
4	43/3	?	?	?	?

The decision for day 4 can already be computed: $\theta_4 = (15+12+16)/3 = 43/3 \simeq 14.33$.

4.3 General result: FTL with squared loss

The previous example suggests a general result: under squared loss, FTL predicts the empirical mean of the past.

Theorem 4.2 (FTL and the empirical mean)**.** *Suppose the observations $y_1, \dots, y_T$ belong to a convex interval $\mathcal{K} \subseteq \mathbb{R}$.*
For the loss

$$\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2,$$

Follow the Leader chooses, for all $t \geq 2$,

$$\boxed{\theta_t = \frac{1}{t-1}\sum_{s=1}^{t-1} y_s}.$$

In other words, the FTL decision at time t is the empirical mean of the previous $t-1$ demand observations.

Proof. By definition,

$$\theta_t \in \arg\min_{\theta \in \mathcal{K}} \left\{ \frac{1}{2}\sum_{s=1}^{t-1}(\theta - y_s)^2 \right\}.$$

Define

$$J_{t-1}(\theta) = \frac{1}{2}\sum_{s=1}^{t-1}(\theta - y_s)^2.$$

Its derivative is

$$J'_{t-1}(\theta) = \sum_{s=1}^{t-1}(\theta - y_s) = (t-1)\theta - \sum_{s=1}^{t-1} y_s.$$

The first-order condition gives

$$J'_{t-1}(\theta_t) = 0 \quad \Longleftrightarrow \quad (t-1)\theta_t - \sum_{s=1}^{t-1} y_s = 0.$$

Hence,

$$\theta_t = \frac{1}{t-1}\sum_{s=1}^{t-1} y_s.$$

Finally,

$$J''_{t-1}(\theta) = t - 1 > 0.$$

The objective is strictly convex. Its solution is therefore the unique global minimizer.
Since all observations lie in the convex interval $\mathcal{K}$, their mean also belongs to $\mathcal{K}$. □

We obtain a recursive update, valid for $t \geq 2$:

$$\boxed{\theta_{t+1} = \theta_t + \frac{1}{t}(y_t - \theta_t).}$$

The weight given to each new observation decreases as $1/t$: the historical average adapts slowly to persistent changes.

Example 4.3 (Why regret is not the same as loss)**.** A large cumulative loss does not necessarily indicate a poor algorithm: the best fixed decision may also incur a large loss if demand is unpredictable. Conversely, a small loss may correspond to large regret if a constant comparator performed much better. Regret therefore *subtracts* a common reference for the observed sequence.

4.4 Computing realized regret

Over the first three days, the learner incurred $12.5 + 4.5 + 3.125 = 20.125$. The best constant selected *in hindsight* is $\theta^\star = (15 + 12 + 16)/3 = 43/3$. Its loss is

$$\min_{\theta\in[0,20]} \sum_{t=1}^{3} \ell_t(\theta) = \frac{1}{2}\left(\frac{4}{9} + \frac{49}{9} + \frac{25}{9}\right) = \frac{13}{3}.$$

Therefore,

$$\boxed{\mathrm{Reg}_3 = 20.125 - \frac{13}{3} = \frac{379}{24} \simeq 15.79.}$$

This regret measures the cost of information unavailable when making the initial decisions; it does not imply that regret grows linearly over longer horizons.

> *Remark* 4.4 (FTL and regime changes). The recursion $\theta_{t+1} = \theta_t + (y_t - \theta_t)/t$ shows that the response of a historical average to a new observation decreases with t. If demand rises suddenly after several stable months, the forecast will continue to reflect the previous regime for a long time. This phenomenon motivates rolling windows, adaptive weighting, and, more generally, algorithms designed for nonstationarity.

4.5 Why a historical average reacts slowly

The recursive form of FTL is also easy to understand under a regime change. Suppose the first demand observations all equal a, then demand suddenly jumps to $b > a$ after day τ:

$$y_s = \begin{cases} a, & s \le \tau, \\ b, & s > \tau. \end{cases}$$

For $t > \tau + 1$, the FTL decision is explicitly

$$\theta_t = \frac{\tau a + (t - 1 - \tau)b}{t - 1} = b - \frac{\tau}{t-1}(b - a).$$

After the change, the error $b - \theta_t$ decreases as $\tau/(t-1)$: the historical mean remains influenced by the τ old observations for a long time. This illustrates the difference between *regret against a constant* and *adaptation to a changing environment*. The former criterion does not require the learner to track every demand change instantly.

5 FTL: Inequalities and Regret Guarantees

5.1 Be-the-Leader: a hypothetical comparison

Let $F_t(\theta) = \sum_{s=1}^{t} \ell_s(\theta)$ and $F_0 \equiv 0$. Thus, $\theta_{t+1} \in \arg\min_{\theta\in\mathcal{K}} F_t(\theta)$. The decision θ_{t+1} already depends on ℓ_t; therefore it is not available to the learner *before* that loss is revealed.

Theorem 5.1 (Be-the-Leader inequality). *For every $u \in \mathcal{K}$, assuming exact FTL minimizers,*

$$\boxed{\sum_{t=1}^{T} \ell_t(\theta_{t+1}) \le \sum_{t=1}^{T} \ell_t(u).}$$

Proof. By definition, $\ell_t(\theta_{t+1}) = F_t(\theta_{t+1}) - F_{t-1}(\theta_{t+1})$. Since θ_t minimizes F_{t-1},

$$\ell_t(\theta_{t+1}) \le F_t(\theta_{t+1}) - F_{t-1}(\theta_t).$$

The sum telescopes: $\sum_t \ell_t(\theta_{t+1}) \le F_T(\theta_{T+1}) - F_0(\theta_1) = F_T(\theta_{T+1}) \le F_T(u)$. □

5.2 Regret decomposition and stability

By adding and subtracting the loss of the updated leader,

$$\mathrm{Reg}_T(u) = \underbrace{\sum_{t=1}^{T}\big[\ell_t(\theta_t) - \ell_t(\theta_{t+1})\big]}_{\text{cost of the one-round delay}} + \underbrace{\sum_{t=1}^{T}\big[\ell_t(\theta_{t+1}) - \ell_t(u)\big]}_{\leq 0 \text{ (Be-the-Leader)}}.$$

Proposition 5.2 (FTL stability bound)**.** *For any comparator $u \in \mathcal{K}$,*

$$\boxed{\mathrm{Reg}_T(u) \leq \sum_{t=1}^{T}\big[\ell_t(\theta_t) - \ell_t(\theta_{t+1})\big].}$$

The difference $\theta_{t+1} - \theta_t$ measures how the algorithm responds to new information. If the minimizers change little, we can hope to achieve small regret.

Intuition behind the cost of delay. Be-the-Leader can minimize the total loss because the decision θ_{t+1} already uses the current round's loss. FTL only uses $\ell_1, \ldots, \ell_{t-1}$ and pays for this missing information. The preceding decomposition shows that it suffices to bound the cost incurred between two consecutive decisions. Strong curvature in the historical objective makes its minimizer more difficult to move.

Example 5.3 (Numerical check of the telescoping identity)**.** In our three-day example, the best decision after the first t days is θ_{t+1}. We can therefore check the regret identity directly using $\Delta_t = F_t(\theta_t) - F_t(\theta_{t+1})$:

t	$F_t(\theta_t)$	$F_t(\theta_{t+1})$	Δ_t
1	25/2	0	25/2
2	9/2	9/4	9/4
3	43/8	13/3	25/24

Summing gives

$$\sum_{t=1}^{3}\Delta_t = \frac{25}{2} + \frac{9}{4} + \frac{25}{24} = \frac{379}{24} = \mathrm{Reg}_3.$$

We recover exactly the regret computed earlier. It is nonnegative for this FTL strategy with exact minimizers, since each Δ_t measures the improvement obtained by reoptimizing F_t.

5.3 Strong convexity: curvature stabilizes the leader

Definition 5.4 (Strong convexity)**.** A convex function f is α-strongly convex over $\mathcal{K}$ ($\alpha > 0$) if, for every $u, v \in \mathcal{K}$ and $g \in \partial f(u)$,

$$\boxed{f(v) \geq f(u) + \langle g, v - u\rangle + \frac{\alpha}{2}\|v - u\|_2^2.}$$

If f is twice differentiable, the condition $\nabla^2 f \succeq \alpha I_d$ is sufficient.

Each loss $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$ is 1-strongly convex, since $\ell_t'' = 1$. If all losses are α-strongly convex,

their sum F_t is $t\alpha$-strongly convex. The curvature of the cumulative objective therefore increases with the number of observations.

Theorem 5.5 (FTL: logarithmic regret)**.** *Suppose $\mathcal{K}$ is convex, exact minimizers exist, each ℓ_t is α-strongly convex ($\alpha > 0$), and there are subgradients $g_t \in \partial\ell_t(\theta_t)$ with $\|g_t\|_2 \le G$ for every t. Then, for any $u \in \mathcal{K}$,*

$$\boxed{\mathrm{Reg}_T(u) \le \frac{G^2}{2\alpha}\sum_{t=1}^{T}\frac{1}{t} \le \frac{G^2}{2\alpha}(1+\log T).}$$

Example 5.6 (Effect of curvature on a minimizer)**.** Consider the family of quadratic functions $H_t(\theta) = \frac{t}{2}(\theta - m_t)^2$. Its curvature is t and its minimum is at m_t. For a fixed displacement δ, the increase in the objective is $H_t(m_t+\delta) - H_t(m_t) = t\delta^2/2$. Thus the same displacement becomes increasingly costly as the number of observations grows. The following proof formalizes this stability without assuming a probabilistic model.

5.4 Proof of the logarithmic bound

The proof explains the role of curvature and why the harmonic series appears. Define

$$\Delta_t := F_t(\theta_t) - F_t(\theta_{t+1}) \ge 0, \quad d_t := \|\theta_t - \theta_{t+1}\|_2.$$

1. Telescoping identity. Since $\ell_t(\theta_t) = F_t(\theta_t) - F_{t-1}(\theta_t)$,

$$\sum_{t=1}^{T}\ell_t(\theta_t) = \sum_{t=1}^{T}\Delta_t + F_T(\theta_{T+1}).$$

Because $F_T(\theta_{T+1}) \le F_T(u)$, $\mathrm{Reg}_T(u) \le \sum_t \Delta_t$; when $u = \theta_{T+1}$, the inequality becomes an equality.

2. Accumulated strong convexity. The point θ_t minimizes F_{t-1}, which is $(t-1)\alpha$-strongly convex; hence,

$$F_{t-1}(\theta_{t+1}) - F_{t-1}(\theta_t) \ge \frac{(t-1)\alpha}{2}d_t^2.$$

Using strong convexity of the current loss and $g_t \in \partial\ell_t(\theta_t)$, we also obtain

$$\ell_t(\theta_{t+1}) - \ell_t(\theta_t) \ge \langle g_t, \theta_{t+1} - \theta_t\rangle + \frac{\alpha}{2}d_t^2.$$

Adding the two inequalities and applying Cauchy–Schwarz gives

$$\Delta_t \le \langle g_t, \theta_t - \theta_{t+1}\rangle - \frac{t\alpha}{2}d_t^2 \le Gd_t - \frac{t\alpha}{2}d_t^2.$$

3. Optimizing the upper bound. The concave function $d \mapsto Gd - t\alpha d^2/2$ reaches its maximum at $d = G/(t\alpha)$. Therefore,

$$\boxed{\Delta_t \le \frac{G^2}{2\alpha t}, \qquad \sum_{t=1}^{T}\Delta_t \le \frac{G^2}{2\alpha}\sum_{t=1}^{T}\frac{1}{t}.}$$

Finally, $\sum_{t=1}^{T} 1/t \le 1 + \log T$. Strong convexity thus causes the per-round contribution to regret to decrease as $1/t$, yielding logarithmic cumulative growth.

Corollary 5.7 (Average regret under the theorem's assumptions)**.** *For standard regret,* $0 \leq \mathrm{Reg}_T \leq \frac{G^2}{2\alpha}(1+\log T)$*, and therefore*

$$\boxed{\frac{\mathrm{Reg}_T}{T} \longrightarrow 0.}$$

Remark 5.8 (The gradient bound is a separate assumption). The squared loss is strongly convex on $\mathbb{R}$, but $\nabla \ell_t(\theta) = \theta - y_t$ is not uniformly bounded over $\mathbb{R}$ if demands and decisions are unbounded. Thus one must separately control curvature α and subgradient magnitude G: neither condition replaces the other.

5.5 Application and limitations of the assumptions

In our demand example, $\mathcal{K} = [0, 20]$, $y_t \in [0, 20]$, and $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$. Thus $\alpha = 1$ and $|\ell_t'(\theta_t)| = |\theta_t - y_t| \leq 20$: we may take $G = 20$. The theorem guarantees, for *every* sequence of bounded demand observations,

$$\boxed{\mathrm{Reg}_T \leq 200(1+\log T).}$$

This is a deliberately conservative bound. It requires neither independence, nor stationarity, nor a probabilistic demand model.

Remark 5.9 (Why FTL can fail). Without strong convexity or a bound on gradient magnitude, the historical decisions may oscillate under changing losses, and sublinear regret is not guaranteed. Stability can be increased through regularization: this is the principle of FTRL (*Follow the Regularized Leader*).

5.6 A counterexample: linear regret without stability

The preceding theorem gives *sufficient conditions*, not a property of all convex losses. We show that FTL can fail badly when the historical objective is unstable.

Example 5.10 (Alternating linear losses)**.** Choose $\mathcal{K} = [-1, 1]$ and

$$\ell_t(\theta) = \begin{cases} \theta, & t \text{ odd}, \\ -\theta, & t \text{ even}. \end{cases}$$

Initialize $\theta_1 = 1$ and choose 1 as the tie-breaking value whenever multiple historical minimizers exist. In round one, the loss $\ell_1(\theta) = \theta$ pushes the next leader to $\theta_2 = -1$. In round two, $\ell_1 + \ell_2 \equiv 0$: under our tie-breaking rule, $\theta_3 = 1$. This mechanism repeats:

$$\boxed{\theta_t = (-1)^{t+1}, \qquad \ell_t(\theta_t) = 1, \quad \forall t.}$$

For every even horizon T, the cumulative loss of any fixed decision is $\sum_{t=1}^{T} \ell_t(u) = 0$ for all $u \in [-1, 1]$. But FTL incurs T units of loss:

$$\boxed{\mathrm{Reg}_T = T, \qquad \frac{\mathrm{Reg}_T}{T} = 1 \quad (T \text{ even}).}$$

FTL constantly follows the past minimizer, but this minimizer switches every round. Regret is linear, not sublinear.

This example contrasts two geometries: linear losses have zero curvature, and their minimizers can jump from one endpoint of $\mathcal{K}$ to the other; strongly convex losses penalize such changes. **FTRL** will add regularization to stabilize decisions when the losses themselves provide insufficient curvature.

Key takeaway

Logical chain: losses accumulate $\Rightarrow$ the curvature of F_t increases as $t\alpha$ $\Rightarrow$ the cost of a change in leader is bounded by $G^2/(2\alpha t)$ $\Rightarrow$ $\mathrm{Reg}_T = O(\log T)$, and average regret becomes negligible.

5.7 Summary of the session

This document has followed a single question, from statistical models to online decisions: *how do we choose a parameter when information arrives progressively?*

Concept	Result and purpose
Empirical risk	Learn a parameter by minimizing observed losses.
Subgradient	Extend optimization to nonsmooth convex losses.
Hindsight	Fixed comparator that knows the entire loss sequence.
Quadratic FTL	Empirical mean of the past; update weight $1/t$.
Be-the-Leader	Control FTL regret through its stability cost.
Strong convexity	Accumulated curvature $t\alpha$; regret contributions proportional to $1/t$.
Logarithmic regret	Under the assumptions: $O(\log T)$, hence negligible average regret.

Remark 5.11 (Interpreting the guarantees). The $O(\log T)$ result is a *conditional* guarantee: it requires, in particular, strong convexity and bounded subgradients. The linear-loss counterexample shows that this conclusion does not follow from the definition of FTL alone. The natural next step is to compare FTL with OGD and FTRL and determine which guarantees remain valid when the losses lack sufficient curvature.